# EEE 339 Power Electronics

## Lecture-by-Lecture Study Guide for the Semester Final (Batch 22)

Built from all 35 lectures in *class slides/*, the 2021 to 2024 finals, the Batch 19/20 term tests, your batch's Assessments #5 to #7, the syllabus and the course outcomes. 34 lecture units, 259 topics.

> **Exam format:** 60 marks. Part A (30) and Part B (30). Each part has 4 questions: Q1 and Q2 are compulsory, then answer Q3 *or* Q4. You write 3 questions per part, about 10 marks each, 6 questions in total.

> **Target: 80 to 90% (48 to 54 marks).** Six questions of about 10 marks. Each lost question costs about 10 marks. So the plan is: (1) every ★★★ lecture fully mastered, which covers all of the compulsory questions we can predict; (2) every ★★ lecture learned well enough for the optional choice; (3) one session on the external zone so a surprise sub-part is still worth partial marks.

## 1. How the paper is set, and why this plan works

- **About 60% comes from the teacher's slides and notes, about 20% from previous papers, about 20% is the external examiner's choice.**
- In the **2024 paper (Batch 21, same teacher, same slides)**, 7 numericals were slide examples copied with the same numbers: Rashid Ex 3.4 (L19), Rashid Ex 10.4 (L22), Hart Ex 3-5 (L17), Hart Ex 4-1 (L18), Hart Ex 6-6 (L13), and both AC-controller class examples (L31_32). About 91% of the paper's marks came from the slides.
- The external examiner's part repeated older papers (the parallel resonant converter came word for word from 2022) or took textbook items next to the slides (for example the SCR gate-pulse-width numerical).
- **Inverters got only 5 marks in 2024**, but your batch had two inverter assessments (#5 and #6). Expect a full inverter question in Part B.
- Topics asked in **all four** years: SCR two-transistor model, cycloconverter, device comparison, full-wave rectifiers, 3-phase rectifiers, buck / buck-boost numericals.

## 2. Part A and Part B: how the lectures are divided

This split follows the 2024 paper (the same teacher). In 2024, Part A held devices and Rashid-style rectifier questions; Part B held Hart DC-DC and rectifier examples, resonant, inverters, AC-AC, cycloconverter and solar. The boundary is not strict, so **L09 (linear regulator vs SMPS), L17 and L18 (Hart rectifier examples) can appear in either part.** Prepare everything; use the split to plan and to balance your time.

### PART A (30 marks): Power Devices + Rectifiers

| Lecture | Title | Priority | Time | When |
|---|---|---|---|---|
| L01 | Introduction to Power Electronics | ★ | 0.25 h | Day 1 |
| L02 | Ideal Switch, Practical Devices and Switch Specifications | ★★ | 0.5 h | Day 1 |
| L03 | Power Semiconductor Switches and Diodes: Classification and Choice | ★★ | 0.5 h | Day 1 |
| L04 | Power Diodes | ★★★ | 0.75 h | Day 1 |

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|---|---|---|---|---|
| L05 | Power BJT | ★★★ | 0.75 h | Day 1 |
| L06 | Power MOSFET and IGBT | ★★★ | 1 h | Day 1 |
| L07 | SCR (Thyristor) | ★★★ | 1.25 h | Day 1 |
| L08 | SCR Dynamic Characteristics and Commutation | ★★★ | 1 h | Day 1 |
| L16 | AC-DC Rectifier 01: Performance Parameters, Single-Phase Half-Wave (R load) | ★★ | 0.75 h | Day 1 |
| L17 | AC-DC Rectifier 02: Half-Wave RL, RL-Source, Freewheeling Diode, Controlled Half-Wave | ★★★ | 1.5 h | Day 1 |
| L18 | AC-DC Rectifier 03: Single-Phase Full-Wave Rectifiers (R, RL, Capacitor Filter) | ★★★ | 1 h | Day 1 |
| L19 | AC-DC Rectifier 04: Full-Wave Rectifier with Highly Inductive Load (Input PF) | ★★★ | 0.75 h | Day 1 |
| L20 | AC-DC Rectifier 05: Controlled Full-Wave Rectifiers and Single-Phase Full Converter | ★★★ | 1.25 h | Day 2 |
| L21 | AC-DC Rectifier 06: Multiphase Star Rectifiers | ★★ | 0.5 h | Day 2 |
| L22 | AC-DC Rectifier 07: Three-Phase Bridge Rectifier and Three-Phase Full Converter | ★★★ | 1.5 h | Day 2 |
| | **Total** | | **13.25 h** | |

## PART B (30 marks): DC-DC, Resonant, Inverters, AC-AC, Cycloconverter, Solar

| Lecture | Title | Priority | Time | When |
|---|---|---|---|---|
| L09 | DC-DC Converter 01: Linear Regulator, Basic Switching Converter, Buck Analysis | ★★★ | 1 h | Day 2 |
| L10 | DC-DC Converter 02: Buck Converter Example and Ripple | ★★★ | 0.5 h | Day 2 |
| L11 | DC-DC Converter 03: Boost Converter | ★★★ | 0.75 h | Day 2 |
| L12 | DC-DC Converter 04: Boost Design 2, Inductor Resistance, Capacitor ESR | ★★ | 0.75 h | Day 2 |
| L13 | DC-DC Converter 05: Buck-Boost Converter | ★★★ | 0.75 h | Day 2 |
| L14 | DC-DC Converter 06: Ćuk Converter | ★★ | 0.75 h | Day 2 |
| L15 | DC-DC Converter 07: SEPIC and Interleaved Converters | ★★★ | 0.75 h | Day 2 |
| L23 | Resonant Converter 01: Zero-Current Switching (ZCS) | ★★ | 0.75 h | Day 3 |
| L24 | Resonant Converter 02: Zero-Voltage Switching (ZVS) and Series Resonant Inverter | ★★ | 0.75 h | Day 3 |
| L25 | Inverter 01: Half-Bridge Inverter, Fourier Analysis, Performance Parameters | ★★★ | 1.25 h | Day 3 |
| L26 | Inverter 02: Single-Phase Full-Bridge Inverter (R, L and RL loads) | ★★★ | 1 h | Day 3 |
| L27 | Problem Solving Class 01 = Assessment #5: Quasi-Square-Wave Full-Bridge Inverter | ★★★ | 0.5 h | Day 3 |
| L28 | Inverter 03: Three-Phase Inverter (180° conduction) | ★★★ | 1.25 h | Day 3 |
| L29 | Inverter 04: Multilevel Inverters (Cascaded H-Bridge) | ★★★ | 1 h | Day 3 |
| L30 | Problem Solving Class 02 = Assessment #6: Three-Phase Inverter, 120° Conduction | ★★★ | 0.5 h | Day 3 |
| L31_32 | AC-AC Converter 01-02: Single-Phase and Three-Phase AC Voltage Controllers | ★★★ | 1.5 h | Day 4 |
| L33 | AC-AC Converter 03: Cycloconverters | ★★★ | 0.75 h | Day 4 |

| | | | | |
|---|---|---|---|---|
| L34 | Problem Solving Class 03 = Assessment #7: Single-Phase AC Voltage Controller | ★★★ | 0.25 h | Day 4 |
| L35 | Solar PV System and Course Wrap-Up | ★★ | 0.5 h | Day 4 |
| | **Total** | | **15.25 h** | |

## 3. How to study each lecture: the 5-step method

1. **Skim (5 min).** Read the lecture's topic list in this guide, then flip through the slide once.
2. **Redraw (15 to 20 min).** Close the slide. Draw every circuit and waveform yourself. Waveforms must be drawn "with proper alignment" (stacked on the same ωt axis). Many marks depend on this.
3. **Derive (15 to 20 min).** Write each derivation from start to boxed result without looking. If you get stuck, look once, then start again from the top.
4. **Solve (20 to 30 min).** Solve every "Do with your own hand" item with the book closed. Compare every number with this guide.
5. **Recall (5 min).** Answer the self-test questions out loud or on paper. Tick the boxes only when you can do it without help. Copy formulas onto your one-page formula sheet.

> **Rules for 4 days:** Do not read passively. Writing beats reading. Study in 50-minute blocks with 10-minute breaks. Use the checklist file (Topic_Checklist.html) to tick off every topic. Sleep at least 6 hours each night; memory consolidates during sleep.

## 4. Four-day timetable

### Day 1: Part A, Devices and Rectifiers I

Lectures: L01 to L08, L16 to L19 · lecture study time 10 h

1. **L01** Introduction to Power Electronics (0.25 h)
2. **L02** Ideal Switch, Practical Devices and Switch Specifications (0.5 h)
3. **L03** Power Semiconductor Switches and Diodes: Classification and Choice (0.5 h)
4. **L04** Power Diodes (0.75 h)
5. **L05** Power BJT (0.75 h)
6. **L06** Power MOSFET and IGBT (1 h)
7. **L07** SCR (Thyristor) (1.25 h)
8. **L08** SCR Dynamic Characteristics and Commutation (1 h)
9. **L16** AC-DC Rectifier 01: Performance Parameters, Single-Phase Half-Wave (R load) (0.75 h)
10. **L17** AC-DC Rectifier 02: Half-Wave RL, RL-Source, Freewheeling Diode, Controlled Half-Wave (1.5 h)
11. **L18** AC-DC Rectifier 03: Single-Phase Full-Wave Rectifiers (R, RL, Capacitor Filter) (1 h)
12. **L19** AC-DC Rectifier 04: Full-Wave Rectifier with Highly Inductive Load (Input PF) (0.75 h)

**Close the day:** End of day: 30 minutes closed-book recall of every self-test question from today.

## Day 2: Part A, Rectifiers II, and Part B, DC-DC

Lectures: L20 to L22, L09 to L15 · lecture study time 8.5 h

1. **L20** AC-DC Rectifier 05: Controlled Full-Wave Rectifiers and Single-Phase Full Converter (1.25 h)
2. **L21** AC-DC Rectifier 06: Multiphase Star Rectifiers (0.5 h)
3. **L22** AC-DC Rectifier 07: Three-Phase Bridge Rectifier and Three-Phase Full Converter (1.5 h)
4. **L09** DC-DC Converter 01: Linear Regulator, Basic Switching Converter, Buck Analysis (1 h)
5. **L10** DC-DC Converter 02: Buck Converter Example and Ripple (0.5 h)
6. **L11** DC-DC Converter 03: Boost Converter (0.75 h)
7. **L12** DC-DC Converter 04: Boost Design 2, Inductor Resistance, Capacitor ESR (0.75 h)
8. **L13** DC-DC Converter 05: Buck-Boost Converter (0.75 h)
9. **L14** DC-DC Converter 06: Ćuk Converter (0.75 h)
10. **L15** DC-DC Converter 07: SEPIC and Interleaved Converters (0.75 h)

**Close the day:** End of day: solve 2024 Part A (Q1 to Q3) under time (75 minutes).

## Day 3: Part B, Resonant and Inverters

Lectures: L23 to L30 · lecture study time 7 h

1. **L23** Resonant Converter 01: Zero-Current Switching (ZCS) (0.75 h)
2. **L24** Resonant Converter 02: Zero-Voltage Switching (ZVS) and Series Resonant Inverter (0.75 h)
3. **L25** Inverter 01: Half-Bridge Inverter, Fourier Analysis, Performance Parameters (1.25 h)
4. **L26** Inverter 02: Single-Phase Full-Bridge Inverter (R, L and RL loads) (1 h)
5. **L27** Problem Solving Class 01 = Assessment #5: Quasi-Square-Wave Full-Bridge Inverter (0.5 h)
6. **L28** Inverter 03: Three-Phase Inverter (180° conduction) (1.25 h)
7. **L29** Inverter 04: Multilevel Inverters (Cascaded H-Bridge) (1 h)
8. **L30** Problem Solving Class 02 = Assessment #6: Three-Phase Inverter, 120° Conduction (0.5 h)

**Close the day:** End of day: redo Assessments #5 and #6 from memory, then 2023 Q5.

## Day 4: Part B, AC-AC and Solar, External Zone, Full Mock

Lectures: L31 to L35, external zone, mock exam · lecture study time 3 h

1. **L31_32** AC-AC Converter 01-02: Single-Phase and Three-Phase AC Voltage Controllers (1.5 h)
2. **L33** AC-AC Converter 03: Cycloconverters (0.75 h)
3. **L34** Problem Solving Class 03 = Assessment #7: Single-Phase AC Voltage Controller (0.25 h)
4. **L35** Solar PV System and Course Wrap-Up (0.5 h)
5. **External zone** (section 6), 1.5 h
6. **Full mock exam**, 6 questions, real time limit
7. **Formula sheet and waveform redraw**, 1.5 h

**Close the day:** Mock: write 6 questions (3 from each part) in the real exam time. Then revise the formula sheet and redraw all key waveforms. Sleep early.

# 5. Lecture-by-lecture guide

## PART A: Power Devices (L01 to L08) and Rectifiers (L16 to L22)

### L01 Introduction to Power Electronics

Part A · ★ Light · 0.25 h · Day 1 · Course overview (no slide content in repo)
Slide: class slides/lec 1 (empty file)

#### TOPICS IN THIS LECTURE (3)

☐ What power electronics is: processing and controlling electric power with switches
☐ Four converter families: AC-DC (rectifier), DC-DC (chopper/regulator), DC-AC (inverter), AC-AC (controller/cycloconverter)
☐ Applications: drives, power supplies, HVDC, renewable energy, EVs

#### DO WITH YOUR OWN HAND

☐ Draw the 4 converter blocks (input to output) from memory

#### PAST-PAPER HITS

- Used only as a lead-in for 'applications' parts of questions

#### HOW TO STUDY IT

1. Read the syllabus page once and memorise the 4 converter types with 2 applications each. 15 minutes.

#### CLOSED-BOOK SELF-TEST (ANSWER WITHOUT LOOKING)

☐ Name the four converter types with one application each

### L02 Ideal Switch, Practical Devices and Switch Specifications

Part A · ★★ Important · 0.5 h · Day 1 · Rashid ch.4 (devices)
Slide: Lecture 02 Ideal Switch.pdf

#### TOPICS IN THIS LECTURE (10)

☐ Ideal switch characteristics: on-state (I_F to infinity, V_ON and R_ON to 0)
☐ Ideal switch characteristics: off-state (V_BR to infinity, I_OFF to 0, R_OFF to infinity)
☐ Ideal switching: t_d, t_r, t_s, t_f to 0; low gate power/voltage/current
☐ Controllable turn-on and turn-off; pulse-only gate signal
☐ High dv/dt and di/dt capability; low thermal impedance; high $i^2t$ (fault current)
☐ Characteristics of practical devices: turn-on/turn-off waveforms, on-state loss P_ON, gate loss P_G
☐ Switching loss P_SW during turn-on and turn-off
☐ Switch specifications: voltage ratings, current ratings (average, rms, repetitive, non-repetitive peak)
☐ Switch specifications: switching speed, di/dt rating, dv/dt rating (junction capacitance), snubbers
☐ Switch specifications: $I^2t$ for fusing, junction/case/storage temperatures, thermal

#### DO WITH YOUR OWN HAND

☐ Write the 10 ideal-switch characteristics in a numbered list from memory

#### PAST-PAPER HITS

- Feeds every 'compare devices / ideal vs practical' question (2021 Q1a, 2023 Q6d)

#### FORMULA SHEET

- `P_total = P_ON + P_SW + P_G (conduction + switching + gate)`

resistance (junction-case, case-sink, sink-ambient)

### HOW TO STUDY IT

1. Read once, then close the slide and write the 10 ideal characteristics.
2. Make a 2-column table: ideal vs practical for each characteristic.
3. Remember dv/dt and di/dt here; you need them again in L07 (SCR turn-on, protection).

### CLOSED-BOOK SELF-TEST (ANSWER WITHOUT LOOKING)

☐ List any 8 characteristics of an ideal switch
☐ What do dv/dt and di/dt ratings mean?

## L03 Power Semiconductor Switches and Diodes: Classification and Choice

Part A · ★★ Important · 0.5 h · Day 1 · Rashid ch.1 and ch.2
Slide: Lecture 03 Power Semiconductor Switches and Diodes.pdf

### TOPICS IN THIS LECTURE (10)

☐ Current-controlled (BJT) vs voltage-controlled (MOSFET) gate drive; why MOS-gate is preferred
☐ Super-power device requirements; SiC devices (lower R_on, ~10x voltage rating)
☐ Classification (9 classes): uncontrolled (diode), controlled-on only (SCR), controlled on and off (BJT, MOSFET, GTO, IGBT, SIT, SITH, MCT)
☐ Classification: continuous gate (BJT, MOSFET, IGBT, SIT) vs pulse gate (SCR, GTO, MCT)
☐ Classification: bipolar-voltage (SCR, GTO) vs unipolar-voltage; bidirectional-current (TRIAC, RCT) vs unidirectional
☐ Device symbols and v-i characteristics (SCR, GTO, MCT, SITH, TRIAC, MOSFET, IGBT)
☐ Control characteristics of power devices (gate signal vs output waveforms)
☐ Device choice: DC input (MOSFET/IGBT) vs AC input (thyristor, TRIAC); MOSFET vs IGBT by voltage/current/frequency
☐ Power diode structure: $n^-$ drift layer, wider depletion, higher V_B
☐ Diode ratings: breakdown voltage V_B, forward voltage V_F, reverse current I_R

### DO WITH YOUR OWN HAND

☐ Make the BJT / MOSFET / IGBT / SCR / GTO / MCT comparison table (power, speed, drive, R_on, blocking voltage)

### PAST-PAPER HITS

- 2021 Q1a (BJT vs MOSFET vs IGBT table, 7 marks)
- 2023 Q6d (BJT, MOSFET, GTO, IGBT, MCT by power and speed)
- 2022 Q4a (BJT vs MOSFET)

### HOW TO STUDY IT

1. This lecture is mainly for one table. Build the comparison table once and keep it on your formula sheet.
2. Speed: MOSFET > IGBT > BJT > GTO > SCR. Power: SCR > GTO > IGBT > BJT > MOSFET.

### CLOSED-BOOK SELF-TEST (ANSWER WITHOUT LOOKING)

☐ Compare BJT, MOSFET, IGBT, GTO, MCT by power capacity and switching speed
☐ Which devices need a continuous gate signal and which need only a pulse?

## L04 Power Diodes

Part A · ★★★ Must master · 0.75 h · Day 1 · Rashid ch.2
Slide: Lecture 04 Power Diodes.pdf

### TOPICS IN THIS LECTURE (10)

### DO WITH YOUR OWN HAND

☐ Dynamic characteristics: intervals t0, t1 (turn-on) and t2, t3, t4 (turn-off)
☐ Reverse recovery time t_rr = t_a + t_b (t3 + t4), peak reverse current I_RR, stored charge Q_RR
☐ Importance of t_rr (decides switching frequency, losses, overvoltage)
☐ Diode types: general-purpose, fast-recovery, Schottky
☐ Common diodes: small-signal, power rectifier, Zener, Schottky, LED, etc.
☐ Schottky diode: metal-semiconductor barrier, low V_F, almost zero t_rr, low reverse voltage
☐ SiC Schottky diodes: features (high voltage, fast, low loss)
☐ Series-connected diodes: voltage sharing with parallel resistors (and capacitors)
☐ Parallel-connected diodes: current sharing with series resistors/inductors
☐ Applications: rectifiers, freewheeling diode, voltage multiplier

☐ Draw the diode turn-off current waveform with t_a, t_b, t_rr, I_RR, Q_RR labelled
☐ Numerical: di/dt = 20 A/μs, t_rr = 5 μs gives Q_RR = 250 μC, I_RR = 100 A

### PAST-PAPER HITS

- 2021 Q2a (significance of t_rr)
- 2021 Q4c and 2023 Q1c (I_RR, Q_RR numerical, same numbers)
- 2023 Q1b (define I_RR, t_rr, importance)
- TT-1 2020 (reverse recovery with diagram)

### FORMULA SHEET

- $Q_{RR} = \frac{1}{2} I_{RR} t_{rr}$
- $I_{RR} = t_a (di/dt) \approx t_{rr} (di/dt)$
- $t_{rr} = \sqrt{(2Q_{RR} / (di/dt))}$
- $I_{RR} = \sqrt{(2Q_{RR} (di/dt))}$

### HOW TO STUDY IT

1. Draw the turn-off waveform 3 times until you can label it without looking.
2. Learn the 4 formulas and do the 20 A/μs numerical. It has appeared 3 times.
3. Write one line each on Schottky, fast-recovery and series/parallel diodes.

### CLOSED-BOOK SELF-TEST (ANSWER WITHOUT LOOKING)

☐ Define t_rr and I_RR and explain their importance
☐ Solve: di/dt = 20 A/μs, t_rr = 5 μs; find I_RR and Q_RR
☐ Why are resistors needed across series-connected diodes?

## L05 Power BJT

Part A · ★★★ Must master · 0.75 h · Day 1 · Rashid ch.4
Slide: Lecture 05 Power BJT.pdf

### TOPICS IN THIS LECTURE (10)

☐ Structure: vertical $n^+$-$n^-$-p-$n^+$, collector drift region (raises blocking voltage)
☐ Basics: large current/power handling, heavily doped emitter, low β, big metal body
☐ Switching operation: saturation (V_CE small) or cutoff; cross the active region fast
☐ Modes: cut-off, active, quasi-saturation, hard saturation
☐ Primary breakdown (avalanche of C-B junction)
☐ Second breakdown: current crowding, hot spots, thermal runaway; how to avoid it
☐ Safe operating area: FBSOA (I_CM, power limit, second-breakdown limit, V_CEO) and RBSOA
☐ Switching characteristics: t_d, t_r (turn-on), t_s, t_f (turn-off); t_on, t_off
☐ Base-drive shaping: overdrive at turn-on, reverse base current to reduce t_s and t_f
☐ Applications of power BJT

### DO WITH YOUR OWN HAND

☐ Draw the BJT switching waveform (i_B, i_C, v_CE) with t_d, t_r, t_s, t_f
☐ Draw the output characteristics showing quasi-saturation and hard saturation

### PAST-PAPER HITS

- 2022 Q4b (switching characteristics of BJT, 6 marks)
- 2024 Q3b (secondary breakdown: why in BJT and thyristor but not MOSFET, 7 marks)
- 2024 Q1d (why BJT/SCR cannot be paralleled but MOSFET can)
- 2023 Q6c (constructional features of power BJT)

### FORMULA SHEET

- $t_{on} = t_d + t_r$
- $t_{off} = t_s + t_f$

### HOW TO STUDY IT

1. Learn 3 diagrams: structure, output characteristics (4 modes), and switching waveform.
2. Write a 6-line answer on second breakdown: cause (negative temperature coefficient of resistance, so current crowds into hot spots), effect, why MOSFETs do not have it (positive temperature coefficient of R_DS(on)).
3. Use the same temperature-coefficient argument for parallel operation: MOSFETs share current naturally; BJTs and SCRs suffer current hogging.

### CLOSED-BOOK SELF-TEST (ANSWER WITHOUT LOOKING)

☐ Explain the switching characteristics of a BJT with waveforms
☐ What is secondary breakdown? Why does it not occur in MOSFETs?
☐ Why can MOSFETs be paralleled easily but BJTs cannot?

## L06 Power MOSFET and IGBT

Part A · ★★★ Must master · 1 h · Day 1 · Rashid ch.4
Slide: Lecture 06 Power MOSFET and IGBT.pdf

### TOPICS IN THIS LECTURE (13)

☐ Power MOSFET types: n/p-channel enhancement, n-channel depletion
☐ VDMOS (double-diffused vertical) structure vs lateral MOSFET; why vertical gives high V and I
☐ Features: voltage-controlled, high input impedance, fast, no second breakdown, unipolar
☐ Disadvantages: high R_DS(on) at high voltage, high conduction loss
☐ Operating regions: cut-off, ohmic (triode), active (saturation); pinch-off condition
☐ Turn-on analysis: intervals (delay, rise, Miller plateau), gate charge
☐ Turn-off analysis: intervals until current falls to zero
☐ MOSFET safe operating area
☐ IGBT: why needed (MOSFET gate plus BJT conduction), conductivity modulation
☐ IGBT structure: $p^+$ injecting layer, $n^-$ drift (n-base), $n^+$ buffer; punch-through vs non-punch-through
☐ IGBT equivalent circuit / half-cell model, operation
☐ IGBT advantages (low on-state drop, voltage drive, high power) and disadvantages (slower than MOSFET, tail current, latch-up)
☐ SCR basic structure (bridge to L07)

### DO WITH YOUR OWN HAND

☐ Draw the VDMOS cross-section
☐ Draw the IGBT cross-section and equivalent circuit
☐ Write a PT vs NPT IGBT comparison (4 points)

### PAST-PAPER HITS

- 2021 Q3d (pinch-off in power MOSFET)
- 2022 Q6b (advantages of IGBT), 2022 Q6d (half-cell model of IGBT)
- 2023 Q3b (IGBT cross-section, PT vs NPT)
- 2022 Q4a (BJT vs MOSFET as a switch)

### FORMULA SHEET

- `MOSFET drain current (active): I_D = K(V_GS − V_T)²`
- `Ohmic region: I_D ≈ 2K(V_GS − V_T)V_DS`

### HOW TO STUDY IT

1. Draw the MOSFET and IGBT cross-sections 2 times each. Label every layer.
2. Make a list of 5 IGBT advantages and 3 disadvantages.
3. Learn the MOSFET regions with the output characteristic curve and the pinch-off line V_DS = V_GS − V_T.

### CLOSED-BOOK SELF-TEST (ANSWER WITHOUT LOOKING)

☐ Draw the cross-section of an IGBT and differentiate PT and NPT IGBTs
☐ Write 4 advantages of IGBT as a switch
☐ Explain pinch-off in a power MOSFET

## L07 SCR (Thyristor)

Part A · ★★★ Must master · 1.25 h · Day 1 · Rashid ch.9
Slide: Lecture 07 SCR.pdf

### TOPICS IN THIS LECTURE (11)

☐ Thyristor family; bistable switch; SCR vs GTO; thyristor vs transistor (losses vs speed)
☐ PNPN structure, three junctions J1, J2, J3, terminals A, K, G
☐ I-V characteristics: forward blocking, forward breakover V_BO, on-state, reverse blocking, reverse breakdown
☐ Latching current I_L and holding current I_H (I_L > I_H)
☐ Turn-on methods: thermal, light, high forward voltage, dv/dt, gate current
☐ Two-transistor model and anode current $I_A = (\alpha_2 I_G + I_{CBO1} + I_{CBO2}) / (1 - (\alpha_1 + \alpha_2))$
☐ Regenerative (latching) action: $\alpha_1 + \alpha_2 \to 1$
☐ Thyristor transient model: junction capacitance; dv/dt turn-on ($i = C_{j2}\, dv/dt$)
☐ Turning off: reduce anode current below I_H; turn-off time $t_q = t_{rr} + t_{rc}$
☐ Forced-commutated thyristor circuit
☐ di/dt protection (series inductor) and dv/dt protection (RC snubber)

### DO WITH YOUR OWN HAND

☐ Derive I_A from the two-transistor model (draw the model first)
☐ Draw the SCR I-V curve with I_L, I_H, V_BO, gate-current family
☐ Gate pulse width numerical (2024 Q1c): $i(t) = (V_s/R)(1 - e^{(-tR/L)})$; $t_G = t(I_H) + t_d \approx 62.2$ µs

### PAST-PAPER HITS

- Two-transistor model in EVERY final: 2021 Q2b, 2022 Q4c, 2023 Q2a, 2024 Q1a
- 2021 TT-1 (five turn-on methods; effect of dv/dt)
- 2020 TT-1 (I-V curve, holding and latching current)
- 2024 Q1c (minimum gate pulse width; this example is not in the slides)

### FORMULA SHEET

- $I_A = (\alpha_2 I_G + I_{CBO1} + I_{CBO2}) / (1 - \alpha_1 - \alpha_2)$
- $t_q = t_{rr} + t_{rc}$
- $i_{j2} = C_{j2}\, (dv/dt)$

### HOW TO STUDY IT

1. This is the single most repeated topic in 4 years. Practise the derivation until you can write it in 6 minutes.
2. Steps: split PNPN into PNP (Q1) and NPN (Q2), then write $I_{C1} = \alpha_1 I_A + I_{CBO1}$ and $I_{C2} = \alpha_2 I_K + I_{CBO2}$, then $I_A = I_{C1} + I_{C2}$ with $I_K = I_A + I_G$, then solve for $I_A$.
3. Learn 5 turn-on methods with one line each, and the I-V curve.
4. Do the gate pulse width numerical once ($V_s = \sqrt{2}\cdot 120 \cdot \sin 30° = 84.85$ V).

### CLOSED-BOOK SELF-TEST (ANSWER WITHOUT LOOKING)

☐ With the two-transistor model, explain the turn-on of an SCR (with equations)
☐ Define latching and holding current on the I-V curve
☐ Name 5 methods to turn on an SCR; explain the dv/dt effect

## L08 SCR Dynamic Characteristics and Commutation

Part A · ★★★ Must master · 1 h · Day 1 · Rashid ch.9 / commutation classes
Slide: Lecture 08 SCR Commutation.pdf

### TOPICS IN THIS LECTURE (11)

☐ Dynamic turn-on: delay time t_d, rise time t_r, spread time t_p; t_on
☐ Dynamic turn-off switching characteristics; reverse recovery time t_rr and gate recovery time t_gr; $t_q = t_{rr} + t_{gr}$
☐ Important point: circuit turn-off time t_c must exceed device turn-off time t_q
☐ Commutation definition; two conditions to turn off an SCR

### DO WITH YOUR OWN HAND

☐ Draw one forced commutation circuit (Class C) with waveforms
☐ Draw the turn-on and turn-off characteristic (i_A, v_AK vs t)

### PAST-PAPER HITS

- 2022 Q4d (methods of SCR turn-on and turn-off, 8 marks)
- 2024 Q1b (natural vs forced commutation with a device example)

☐ Natural (line / Class F) commutation: AC supply, rectifiers, AC controllers, cycloconverters
☐ Forced commutation: needed in DC circuits (choppers, inverters)
☐ Class A: self/resonant/load commutation (underdamped RLC load)
☐ Class B: resonant LC across the SCR (self commutation)
☐ Class C: complementary commutation (main SCR + auxiliary SCR + capacitor)
☐ Class D: auxiliary (impulse) commutation
☐ Class E: external pulse commutation

- 2021 Q3c (describe one forced commutation circuit with waveforms)

### FORMULA SHEET

- `t_on = t_d + t_r + t_p`
- `t_q = t_rr + t_gr`
- `Condition: t_c > t_q`

### HOW TO STUDY IT

1. Learn natural vs forced as a 2-column table (source of reverse voltage, AC vs DC, examples).
2. Learn Class A to E with one line each. Pick Class C and learn its full circuit and waveforms. This is enough for the 'describe any one' question.
3. Example for natural commutation: SCR in an AC phase-controlled rectifier. For forced: SCR in a DC chopper.

### CLOSED-BOOK SELF-TEST (ANSWER WITHOUT LOOKING)

☐ Explain natural and forced commutation with one example each
☐ Describe Class C commutation with circuit and waveforms
☐ Why must t_c be greater than t_q?

## L16 AC-DC Rectifier 01: Performance Parameters, Single-Phase Half-Wave (R load)

Part A · ★★ Important · 0.75 h · Day 1 · Rashid (old ed.) ch.2: 2-5 to 2-8
Slide: Lecture 16 AC-DC Rectifier 01.pdf

### TOPICS IN THIS LECTURE (7)

☐ Performance parameters: $V_{dc}$, $I_{dc}$, $P_{dc}$, $V_{rms}$, $I_{rms}$, $P_{ac}$
☐ Efficiency $\eta = P_{dc}/P_{ac}$; form factor $FF = V_{rms}/V_{dc}$; ripple factor $RF = \sqrt{(FF^2 - 1)}$
☐ Transformer utilization factor $TUF = P_{dc}/(V_s I_s)$; peak inverse voltage PIV
☐ Displacement factor, harmonic factor HF, input power factor PF, crest factor
☐ Single-phase half-wave rectifier with R load: waveforms and analysis
☐ Results: $\eta = 40.5\%$, $FF = 157\%$, $RF = 121\%$, $TUF = 0.286$, $PIV = V_m$
☐ DC saturation problem of the transformer core in half-wave rectifiers

### DO WITH YOUR OWN HAND

☐ Derive $V_{dc} = V_m/\pi$ and $V_{rms} = V_m/2$, then compute η, FF, RF, TUF for the half-wave R rectifier

### PAST-PAPER HITS

- Definitions support every rectifier numerical (TUF and efficiency asked in 2024 Q3c)

### FORMULA SHEET

- `V_dc = 0.318V_m, V_rms = 0.5V_m (half-wave R)`
- `η = P_dc/P_ac`
- `FF = V_rms/V_dc`
- `RF = √(FF² − 1)`
- `TUF = P_dc/(V_s I_s)`
- `HF = √((I_s/I_s1)² − 1)`
- `PF = (I_s1/I_s) cos φ`

### HOW TO STUDY IT

1. Learn the 8 definitions as formulas. They are used in L19, L20 and L22.
2. Do the half-wave numbers yourself once (η = 40.5%, RF = 121%, TUF = 0.286).

### CLOSED-BOOK SELF-TEST (ANSWER WITHOUT LOOKING)

☐ Define FF, RF, TUF, HF, DF and PF
☐ Find η, FF, RF, TUF and PIV of a single-phase half-wave rectifier with R load

## L17 AC-DC Rectifier 02: Half-Wave RL, RL-Source, Freewheeling Diode, Controlled Half-Wave

Part A · ★★★ Must master · 1.5 h · Day 1 · Hart ch.3: 3.3, 3.5, 3.6, 3.7, 3.9

Slide: Lecture 17 AC-DC Rectifier 02.pdf

### TOPICS IN THIS LECTURE (8)

☐ Half-wave rectifier with RL load: current = forced + natural response; extinction angle β

☐ Average current, rms current, power and PF for RL load

☐ RL-source load (battery charging): α = $\sin^{-1}(V_{dc}/V_m)$, current expression, $P_R$, $P_{dc}$, PF

☐ Inductor-source load

☐ Freewheeling diode: purpose, operation, waveforms of $v_o$, $i_{D1}$, $i_{D2}$

☐ Freewheeling diode with large L ($L/R \to \infty$): constant load current

☐ Controlled half-wave rectifier (SCR) with R load: $V_o = (V_m/2\pi)(1 + \cos\alpha)$

☐ Controlled half-wave rectifier with RL load

### DO WITH YOUR OWN HAND

☐ Ex 3-2 ($R = 100\ \Omega$, $L = 0.1$ H, $V_m = 100$ V): $P = 22.4$ W, pf = 0.67

☐ Ex 3-5 RL-source ($R = 2\ \Omega$, $L = 20$ mH, $V_{dc} = 100$ V, 120 V): $P_R \approx 31.7$ W, $P_{dc} = 225$ W, pf ≈ 0.54

☐ Ex 3-7 freewheeling ($R = 2\ \Omega$, $L = 25$ mH, $V_m = 100$ V): $V_o = 31.8$ V, $I_o = 15.9$ A

☐ Ex 3-8 freewheeling, $L \to \infty$ (240 V, $R = 8\ \Omega$): $V_o = 108$ V, $I_o = 13.5$ A, $P \approx 1459$ W, pf ≈ 0.64

☐ Ex 3-10 controlled half-wave (40 V across 100 Ω from 120 V): $\alpha = 61.2°$, $P = 57.2$ W, pf = 0.63

☐ Derive $i(\omega t)$, $I_{avg}$, $I_{rms}$, P and PF for a half-wave RL-source load (2022 Q1b)

### PAST-PAPER HITS

- 2024 Q4b = Ex 3-5 verbatim (10 marks)
- 2022 Q1b,c (RL-source derivation; half-wave RL numerical $R = 25\ \Omega$, $L = 150$ mH)
- 2023 Q2b (controlled half-wave RL: 120 V, $R = 20\ \Omega$, $L = 0.04$ H, $\alpha = 45°$)
- 2022 Q1a(ii) (short note: freewheeling diode)

### FORMULA SHEET

- $i(\omega t) = (V_m/Z)[\sin(\omega t - \theta) + \sin\theta \cdot e^{(-\omega t/\omega\tau)}]$, $Z = \sqrt{(R^2 + (\omega L)^2)}$, $\theta = \tan^{-1}(\omega L/R)$, $\omega\tau = \omega L/R$
- RL-source: $i(\omega t) = (V_m/Z)\sin(\omega t - \theta) - V_{dc}/R + A \cdot e^{(-\omega t/\omega\tau)}$, from α to β
- $\alpha = \sin^{-1}(V_{dc}/V_m)$; $P_{dc} = I_o V_{dc}$; $P_R = I_{rms}^2 R$
- Freewheeling: $V_o = V_m/\pi$; $I_o = V_o/R$
- Controlled half-wave R: $V_o = (V_m/2\pi)(1 + \cos\alpha)$

### HOW TO STUDY IT

1. Work through each Hart section in order: RL (3.3), RL-source (3.5), freewheeling (3.7), controlled (3.9).
2. For each: draw the circuit and $v_o$, $i_o$ waveforms, then write the current equation, then solve the example with the book closed.
3. Ex 3-5 came verbatim in 2024. Practise it twice: find α, then the current expression, then β (numerically), then $I_o$, $I_{rms}$, $P_R$, $P_{dc}$, pf.
4. Use a calculator table for β: try values until $i(\beta) = 0$.

### CLOSED-BOOK SELF-TEST (ANSWER WITHOUT LOOKING)

☐ Derive the current expression for a half-wave rectifier with RL load

☐ Solve Ex 3-5 without the book

☐ Why is a freewheeling diode used? Sketch $v_o$, $i_{D1}$, $i_{D2}$

## L18 AC-DC Rectifier 03: Single-Phase Full-Wave Rectifiers (R, RL, Capacitor Filter)

Part A · ★★★ Must master · 1 h · Day 1 · Hart ch.4: 4.1, 4.2

Slide: Lecture 18 AC-DC Rectifier 03.pdf

### TOPICS IN THIS LECTURE (7)

☐ Bridge and centre-tapped full-wave rectifiers: circuits and operation
☐ Resistive load: $V_o = 2V_m/\pi$, waveforms
☐ RL load: Fourier series of output voltage, harmonic currents, average and rms current
☐ Source current: harmonics, THD and power factor
☐ Diode average and rms currents
☐ Capacitor-filtered full-wave rectifier: ripple $\Delta V_o \approx V_m/(2fRC)$
☐ Half-wave capacitor filter reference (Hart Ex 3-9): $\Delta V_o \approx V_m/(fRC)$

### DO WITH YOUR OWN HAND

☐ Ex 4-1 ($V_m$ = 100 V, R = 10 Ω, L = 10 mH): $I_o$ = 6.37 A, $\Delta I_{pp} \approx$ 6.8 A, P = 464 W, pf = 0.964, $I_{D,avg}$ = 3.18 A, $I_{D,rms}$ = 4.82 A
☐ Ex 4-4 capacitor filter (120 V, R = 500 Ω, C = 100 µF): $\Delta V_o \approx$ 28.3 V; C for 1% ripple ≈ 1667 µF
☐ Draw $v_s$, $v_o$, $i_o$, $i_{D1,2}$, $i_{D3,4}$, $i_s$ of a bridge with RL load with proper alignment

### PAST-PAPER HITS

- 2024 Q4c = Ex 4-1 verbatim (10 marks)
- 2020 TT-1 (waveforms of a full-wave bridge with RL load)
- 2021 Q2c (half-wave capacitor filter numerical: 120 V, 50 W, ripple 1.5 V)

### FORMULA SHEET

- $V_o = 2V_m/\pi$
- $V_n = (2V_m/\pi)[1/(n-1) - 1/(n+1)]$, n = 2, 4, 6, …
- $I_n = V_n/Z_n$, $Z_n = |R + jn\omega L|$
- $I_{rms} = \sqrt{(I_o^2 + \Sigma(I_n/\sqrt{2})^2)}$
- Full-wave C filter: $\Delta V_o \approx V_m/(2fRC)$; half-wave: $V_m/(fRC)$

### HOW TO STUDY IT

1. Learn the Fourier method: $V_o$, then $V_2$, $V_4$, then $I_2$, $I_4$, then $I_{rms}$, then P.
2. Solve Ex 4-1 twice (it was asked verbatim in 2024). Then Ex 4-4.
3. Practise one 'proper alignment' waveform drawing: all waveforms stacked on the same ωt axis.

### CLOSED-BOOK SELF-TEST (ANSWER WITHOUT LOOKING)

☐ Solve Ex 4-1 without the book
☐ Derive the capacitor-filter ripple formula
☐ Draw the bridge rectifier waveforms with RL load

## L19 AC-DC Rectifier 04: Full-Wave Rectifier with Highly Inductive Load (Input PF)

Part A · ★★★ Must master · 0.75 h · Day 1 · Rashid 3.5, Example 3.4
Slide: Lecture 19 AC-DC Rectifier 04.pdf

### TOPICS IN THIS LECTURE (5)

☐ Single-phase bridge with a highly inductive load (DC motor): ripple-free $I_a$
☐ Input current is a square wave of amplitude $I_a$
☐ Fourier series of input current: $i_s = \Sigma (4I_a/n\pi) \sin n\omega t$ (n odd)
☐ Fundamental: $I_{s1} = 0.90I_a$; rms $I_s = I_a$
☐ Harmonic factor HF = 48.34%, displacement factor DF = 1, PF = 0.90

### DO WITH YOUR OWN HAND

☐ Ex 3.4: derive the Fourier series of $i_s$, then HF = 48.34% and PF = 0.90 (lagging)

### PAST-PAPER HITS

- 2024 Q2a = Ex 3.4 verbatim (12 marks)

### FORMULA SHEET

- $I_{s1} = 2\sqrt{2}\, I_a/\pi = 0.9003I_a$
- $HF = \sqrt{((I_s/I_{s1})^2 - 1)} = 0.4834$
- $PF = (I_{s1}/I_s)\cdot\cos\varphi = 0.90$

### HOW TO STUDY IT

1. Short lecture but a 12-mark question. Learn the Fourier integral for the square wave: $b_n = 4I_a/(n\pi)$.
2. Write the full solution once, neatly, as you would in the exam.

### CLOSED-BOOK SELF-TEST (ANSWER WITHOUT LOOKING)

☐ Single-phase bridge supplying a DC motor with ripple-free current: find HF and input PF

## L20 AC-DC Rectifier 05: Controlled Full-Wave Rectifiers and Single-Phase Full Converter

Part A · ★★★ Must master · 1.25 h · Day 2 · Hart 4.3; Rashid 10.2
Slide: Lecture 20 AC-DC Rectifier 05.pdf

### TOPICS IN THIS LECTURE (6)

☐ Controlled full-wave bridge (4 SCRs) with R load: $V_o = (V_m/\pi)(1 + \cos\alpha)$
☐ Controlled bridge with RL load: discontinuous current ($\beta < \pi + \alpha$)
☐ Continuous current: $V_o = (2V_m/\pi)\cos\alpha$; condition for continuous current
☐ Single-phase full converter: rectification ($\alpha < 90°$) and inversion ($\alpha > 90°$) modes
☐ Full converter input current Fourier series; HF, DF = $\cos\alpha$, PF
☐ Waveforms: $v_s$, gate pulses, $v_o$, $i_o$, $i_s$

### DO WITH YOUR OWN HAND

☐ Ex 4-6 (120 V, R = 20 Ω, $\alpha = 40°$): $V_o = 95.4$ V, $I_o = 4.77$ A, P ≈ 673 W
☐ Ex 4-7 discontinuous (120 V, R = 10 Ω, L = 20 mH, $\alpha = 60°$): P ≈ 697 W
☐ Ex 10.1 full converter ($\alpha = 60°$): $V_{dc} = 54.02$ V, HF = 48.34%, DF = 0.5, PF = 0.45
☐ Explain the continuous / discontinuous current condition for a controlled bridge with RL load

### PAST-PAPER HITS

- 2022 Q6c and 2023 Q2c = Ex 4-6 (asked twice)
- 2022 Q6a (continuous and discontinuous conditions, 8 marks)
- 2021 Q2d (controlled full-wave with $V_{dc}$ source: find α for 1000 W and L)

### FORMULA SHEET

- R load: $V_o = (V_m/\pi)(1 + \cos\alpha)$; $V_{rms} = (V_m/\sqrt{2})\sqrt{(1 - \alpha/\pi + \sin 2\alpha/2\pi)}$
- Continuous: $V_o = (2V_m/\pi)\cos\alpha$
- Continuous-current condition: $\alpha \le \tan^{-1}(\omega L/R) = \theta$ (i.e. $\beta \ge \pi + \alpha$)
- Full converter: $I_{s1} = 0.90 I_a$, HF = 48.34%, DF = $\cos\alpha$, PF = $0.9\cos\alpha$

### HOW TO STUDY IT

1. Draw the controlled bridge waveforms for R load and for continuous RL load side by side, and see where the negative part of $v_o$ comes from.
2. Solve Ex 4-6 (repeated twice) and Ex 10.1. Ex 10.1 is the partner of Ex 3.4 that was used in 2024.
3. Learn the continuous-current condition as a short paragraph with a sketch.

### CLOSED-BOOK SELF-TEST (ANSWER WITHOUT LOOKING)

☐ Solve Ex 4-6
☐ For a controlled bridge with RL load, when is current continuous?
☐ Solve Ex 10.1 part (b)

## L21 AC-DC Rectifier 06: Multiphase Star Rectifiers

Part A · ★★ Important · 0.5 h · Day 2 · Rashid (old ed.) 2-9
Slide: Lecture 21 AC-DC Rectifier 06.pdf

### TOPICS IN THIS LECTURE (5)

☐ Why multiphase: higher $V_{dc}$, lower ripple, higher ripple frequency
☐ q-phase star rectifier: each diode conducts for $2\pi/q$
☐ Average output voltage $V_{dc} = V_m (q/\pi) \sin(\pi/q)$

### DO WITH YOUR OWN HAND

☐ Derive $V_{dc}$ for a q-phase star rectifier; evaluate for q = 3 ($0.827V_m$) and q = 6 ($0.955V_m$)

### PAST-PAPER HITS

- Background for 3-phase rectifier questions (2023 Q3a)

☐ RMS output voltage and Fourier series of output voltage (harmonics at nq)
☐ Three-phase half-wave (q = 3) and six-phase cases

### FORMULA SHEET

- $V_{dc} = V_m \frac{q}{\pi} \sin\frac{\pi}{q}$
- $V_{rms} = V_m \sqrt{\frac{q}{2\pi}\left(\frac{\pi}{q} + \frac{1}{2}\sin\frac{2\pi}{q}\right)}$

### HOW TO STUDY IT

1. Do the $V_{dc}$ derivation once (integrate cos from $-\pi/q$ to $\pi/q$). 30 minutes is enough.

### CLOSED-BOOK SELF-TEST (ANSWER WITHOUT LOOKING)

☐ Derive $V_{dc}$ for a q-phase rectifier

## L22 AC-DC Rectifier 07: Three-Phase Bridge Rectifier and Three-Phase Full Converter

Part A · ★★★ Must master · 1.5 h · Day 2 · Rashid 3.7, 10.4
Slide: Lecture 22 AC-DC Rectifier 07.pdf

### TOPICS IN THIS LECTURE (9)

☐ Three-phase uncontrolled bridge: circuit, diode conduction order (1-6), 6-pulse output
☐ Waveforms: phase voltages, line voltages, output voltage, diode currents, line currents (proper alignment)
☐ $V_{dc} = 3\sqrt{3}\,V_m/\pi = 1.654V_m$ ($V_m$ = phase peak); $V_{rms} = 1.6554V_m$
☐ 12-pulse rectifier note (Y and Δ transformers in series)
☐ Three-phase full converter: 6 SCRs, gate pulses 60° apart
☐ Derivation of $V_{dc} = (3\sqrt{3}\,V_m/\pi)\cos\alpha$ (= $1.35V_{LL}\cos\alpha$)
☐ $V_{rms}$ of full converter; thyristor average and rms currents
☐ Input current Fourier series: HF = 31.08%, DF = $\cos\alpha$, PF = $0.955\cos\alpha$
☐ Applications: industrial drives up to 120 kW, HVDC

### DO WITH YOUR OWN HAND

☐ Ex 3.7 three-phase bridge ($I_{dc}$ = 60 A, $V_{dc}$ = 280.7 V): η, FF, RF, TUF, PIV = 293.9 V
☐ Derive $V_{dc}(\alpha)$ for the three-phase full converter with waveforms (2024 Q2b, 13 marks)
☐ Ex 10.4 (208 V, R = 10 Ω, 50% of max): α = 60°, $I_{dc}$ = 14.05 A, $I_{rms}$ = 15.93 A, $I_A$ = 4.68 A, $I_R$ = 9.2 A, η = 77.8%, TUF = 0.421, PF = 0.542
☐ Ex 10.5 input PF: HF = 31.08%, DF = $\cos\alpha$, PF = $0.955\cos\alpha$; at α = 60°, PF = 0.478

### PAST-PAPER HITS

- 2024 Q2b (derive $V_{dc}(\alpha)$, 13 marks) and 2024 Q3c = Ex 10.4 verbatim (12 marks)
- 2023 Q3a (3-phase uncontrolled waveforms), 2023 Q3d (3-phase controlled, find α for 50 A, harmonics)
- 2021 Q3b (3-phase converter as inverter, 300 V source)
- 2020 TT-2 (3-phase rectifier 480 V, RL load: rms currents)
- 2023 Q3c (advantages of HVDC)

### FORMULA SHEET

- Uncontrolled: $V_{dc} = 3\sqrt{3}V_m/\pi = 1.654V_m = 1.35V_{LL}$
- Full converter: $V_{dc} = (3\sqrt{3}V_m/\pi)\cos\alpha$
- $V_{rms} = \sqrt{3}V_m\sqrt{\frac{1}{2} + (3\sqrt{3}/4\pi)\cos 2\alpha}$
- $I_A = I_{dc}/3$; $I_R = I_{rms}\sqrt{(2/6)}$; $I_s = I_{rms}\sqrt{(4/6)}$
- Input: HF = 31.08%, DF = $\cos\alpha$, PF = $0.955\cos\alpha$

### HOW TO STUDY IT

1. Draw the 3-phase voltage grid ($v_{an}$, $v_{bn}$, $v_{cn}$ and the 6 line voltages) until you can do it quickly. Every 3-phase question needs it.
2. Learn the derivation: output follows the line voltage $v_{ab}$ from $\pi/6 + \alpha$ to $\pi/2 + \alpha$, so $V_{dc} = (3/\pi)\int\sqrt{3}V_m \sin(\omega t + \pi/6)\, d\omega t$.
3. Solve Ex 10.4 twice; it came verbatim in 2024. Then Ex 10.5.

### CLOSED-BOOK SELF-TEST (ANSWER WITHOUT LOOKING)

- [ ] Derive V_dc(α) for a three-phase full converter with gate-pulse and output waveforms
- [ ] Solve Ex 10.4 without the book
- [ ] Draw the 3-phase bridge rectifier waveforms with proper alignment

# PART B: DC-DC (L09 to L15), Resonant (L23 to L24), Inverters (L25 to L30), AC-AC and Cycloconverter (L31 to L34), Solar (L35)

## L09 DC-DC Converter 01: Linear Regulator, Basic Switching Converter, Buck Analysis

Part B · ★★★ Must master · 1 h · Day 2 · Hart 6.1 to 6.3
Slide: Lecture 09 DC-DC Converter 01.pdf + Lecture 09 Handout.pdf

### TOPICS IN THIS LECTURE (9)

☐ Linear voltage regulator: circuit (transistor in active region), efficiency ≈ $V_o/V_s$
☐ Drawback of linear regulator (high loss when $V_s \gg V_o$) and how switching solves it
☐ Basic switching converter: switch + diode + LC filter, duty ratio D
☐ Buck converter circuit; switch-closed and switch-open equivalent circuits
☐ Inductor voltage/current analysis; volt-second balance gives $V_o = DV_s$
☐ Inductor current: average $I_L = V_o/R$; $I_{max}$, $I_{min}$; $\Delta i_L$
☐ Minimum L for continuous current: $L_{min} = (1 - D)R/(2f)$
☐ Output voltage ripple $\Delta V_o/V_o = (1 - D)/(8LCf^2)$
☐ Handout: fill-in table of the buck equivalent circuits

### DO WITH YOUR OWN HAND

☐ Derive $V_o = DV_s$ for the buck converter from the ON/OFF equivalent circuits
☐ Draw $v_L$, $i_L$, $i_C$, $i_D$ waveforms for the buck converter
☐ Derive the relation between inductor current and diode current from the waveforms (2023 Q4b)

### PAST-PAPER HITS

- 2023 Q4a (linear regulator, drawback, solution)
- 2024 Q3a (DC-DC converters vs switching-mode regulators)
- 2023 Q4b (buck waveforms and relation between $i_L$ and $i_D$)

### FORMULA SHEET

- $V_o = DV_s$
- $\Delta i_L = (V_s - V_o)DT/L = V_o(1 - D)/(Lf)$
- $I_{max,min} = V_o/R \pm \Delta i_L/2$
- $L_{min} = (1 - D)R/(2f)$
- $\Delta V_o/V_o = (1 - D)/(8LCf^2)$

### HOW TO STUDY IT

1. Fill in the handout yourself. It is the derivation skeleton.
2. Learn the 5-step method you will reuse for every converter: (1) ON circuit, $v_L$; (2) OFF circuit, $v_L$; (3) volt-second balance gives $V_o$; (4) power balance or KCL gives $I_L$; (5) $\Delta i_L$ and capacitor charge $\Delta Q$ give the ripple.

### CLOSED-BOOK SELF-TEST (ANSWER WITHOUT LOOKING)

☐ Draw a linear regulator; give its drawback and how to overcome it
☐ Derive $V_o$, $I_{max}$, $I_{min}$, $L_{min}$ and the ripple for a buck converter

## L10 DC-DC Converter 02: Buck Converter Example and Ripple

Part B · ★★★ Must master · 0.5 h · Day 2 · Hart 6.3, Example 6-1
Slide: Lecture 10 DC-DC Converter 02.pdf

### TOPICS IN THIS LECTURE (3)

☐ Output voltage ripple derivation (capacitor charge $\Delta Q = T\Delta i_L/8$)
☐ Example 6-1: buck converter analysis
☐ Design notes: higher f gives smaller L and C

### DO WITH YOUR OWN HAND

☐ Ex 6-1 ($V_s$ = 50 V, D = 0.4, L = 400 µH, C = 100 µF, f = 20 kHz, R = 20 Ω): $V_o$ = 20 V, $I_L$ = 1 A, $I_{max}$ = 1.75 A, $I_{min}$ = 0.25 A, $\Delta V_o/V_o$ = 0.47%

### PAST-PAPER HITS

- 2020 TT-2 Q2 = Ex 6-1 verbatim
- 2022 Q2c (buck design 5 V to 3.3 V, 1 to 2 A; the method is the same)

### FORMULA SHEET

- $\Delta Q = (T/8)\Delta i_L$
- $\Delta V_o = \Delta Q/C$

### HOW TO STUDY IT

1. Solve Ex 6-1 with the book closed and check every number. Then attempt the 2022 buck design question using the same steps.

### CLOSED-BOOK SELF-TEST (ANSWER WITHOUT LOOKING)

☐ Solve Ex 6-1 without the book

## L11 DC-DC Converter 03: Boost Converter

Part B · ★★★ Must master · 0.75 h · Day 2 · Hart 6.5
Slide: Lecture 11 DC-DC Converter 03.pdf

### TOPICS IN THIS LECTURE (8)

☐ Boost circuit; switch-closed and switch-open equivalent circuits
☐ $V_o = V_s/(1 - D)$ from volt-second balance
☐ Average inductor current $I_L = V_s/((1 - D)^2R)$
☐ $I_{max}$, $I_{min}$, $\Delta i_L = V_sDT/L$
☐ Minimum L: $L_{min} = D(1 - D)^2R/(2f)$
☐ Output ripple $\Delta V_o/V_o = D/(RCf)$
☐ Waveforms: $i_L$, $i_D$, $i_C$, $v_o$ (proper alignment)
☐ Example 6-4: boost converter design 1 (12 V to 30 V, 50 Ω, 1% ripple)

### DO WITH YOUR OWN HAND

☐ Full boost derivation: equivalent circuits, waveforms, $V_o/V_s$, $I_L$, $i_L$ vs $i_D$ relation (10 marks in 2021)
☐ Ex 6-4 design: D = 0.6, choose f, L > $L_{min}$, C from $\Delta V_o/V_o$

### PAST-PAPER HITS

- 2021 Q4a (full boost derivation, 10 marks)
- 2023 Q4c (boost: duty, $L_{min}$, $C_{min}$)
- 2020 TT-2 Q1 (boost circuit and waveforms)
- 2019 TT-2 (boost numerical)

### FORMULA SHEET

- $V_o = V_s/(1 - D)$
- $I_L = V_s/((1 - D)^2R)$
- $\Delta i_L = V_sDT/L$
- $L_{min} = D(1 - D)^2R/(2f)$
- $\Delta V_o/V_o = D/(RCf)$

### HOW TO STUDY IT

1. Repeat the 5-step method from L09 for the boost converter. Then do the 2023 Q4c numerical (25 W at 15 V from 6 V) as practice.

### CLOSED-BOOK SELF-TEST (ANSWER WITHOUT LOOKING)

☐ Derive $V_o/V_s$ for a boost converter with equivalent circuits and waveforms
☐ Solve Ex 6-4

## L12 DC-DC Converter 04: Boost Design 2, Inductor Resistance, Capacitor ESR

Part B · ★★ Important · 0.75 h · Day 2 · Hart 6.5
Slide: Lecture 12 DC-DC Converter 04.pdf

### TOPICS IN THIS LECTURE (5)

☐ Example 6-4 continued (component selection)
☐ Example 6-5: boost design 2 (8 V, 1 A out; 2.7 to 4.2 V in; $\Delta i_L \leq 40\%$)
☐ Effect of capacitor ESR on output ripple: $\Delta V_{o,ESR} = \Delta i_C \cdot r_C$
☐ Effect of inductor resistance on boost output: $V_o = (V_s/(1 - D)) \cdot (1/(1 + r_L/(R(1 - D)^2)))$
☐ Efficiency drop and maximum achievable gain because of $r_L$

### DO WITH YOUR OWN HAND

☐ Ex 6-5: design for the worst-case input
☐ Explain how ESR and inductor resistance affect DC-DC output (2022 Q2b, 7 marks)

### PAST-PAPER HITS

- 2022 Q2b (capacitor and inductor resistance effects, 7 marks)
- 2022 Q2c (maximum ESR of capacitor)

### FORMULA SHEET

- $\Delta V_{o,ESR} = \Delta i_C r_C$
- Boost with $r_L$: $V_o = V_s/(1 - D) \times 1/(1 + r_L/(R(1 - D)^2))$

### HOW TO STUDY IT

1. Learn the ESR and $r_L$ paragraphs with one formula each; sketch $V_o/V_s$ vs D for ideal and with $r_L$ (the gain curve bends down). Do Ex 6-5 once.

### CLOSED-BOOK SELF-TEST (ANSWER WITHOUT LOOKING)

☐ How do ESR and inductor resistance affect the DC-DC converter output?

## L13 DC-DC Converter 05: Buck-Boost Converter

Part B · ★★★ Must master · 0.75 h · Day 2 · Hart 6.6

Slide: Lecture 13 DC-DC Converter 05.pdf

### TOPICS IN THIS LECTURE (6)

☐ Buck-boost circuit; ON and OFF equivalent circuits; output polarity is inverted
☐ $V_o = -V_s D/(1 - D)$
☐ $I_L = V_sD/(R(1 - D)^2)$; $I_{max}$, $I_{min}$; $\Delta i_L = V_sDT/L$
☐ $L_{min} = (1 - D)^2R/(2f)$
☐ Output ripple $\Delta V_o/V_o = D/(RCf)$; ESR effect
☐ Waveforms: $i_L$, $i_D$, $i_C$, $v_o$

### DO WITH YOUR OWN HAND

☐ Ex 6-6 (24 V, D = 0.4, R = 5 Ω, L = 20 µH, C = 80 µF, 100 kHz): $V_o = -16$ V, $I_L = 5.33$ A, $I_{max} = 7.73$ A, $I_{min} = 2.93$ A, $\Delta V_o/V_o = 1\%$
☐ Derive $V_o$, $I_L$, $L_{min}$ and $\Delta V_o$ for a buck-boost (2022 Q2a, 8 marks)
☐ 2021 Q4b (24 V to −36 V, 10 Ω, 100 kHz: L for $I_{min}$ = 40% of average, C for 0.5% ripple)

### PAST-PAPER HITS

- 2024 Q5b = Ex 6-6 verbatim
- 2022 Q2a (buck-boost derivation)
- 2021 Q4b (buck-boost design)
- 2019 TT-2 (buck-boost circuit and waveforms)

### FORMULA SHEET

- $V_o = -V_sD/(1 - D)$
- $I_L = V_sD/(R(1 - D)^2)$
- $L_{min} = (1 - D)^2R/(2f)$
- $\Delta V_o/V_o = D/(RCf)$

### HOW TO STUDY IT

1. Same 5-step method. Solve Ex 6-6 and 2021 Q4b. Note the inverted output in every waveform.

### CLOSED-BOOK SELF-TEST (ANSWER WITHOUT LOOKING)

☐ Derive the buck-boost output voltage and minimum inductance
☐ Solve Ex 6-6 without the book

## L14 DC-DC Converter 06: Ćuk Converter

Part B · ★★ Important · 0.75 h · Day 2 · Hart 6.7

Slide: Lecture 14 DC-DC Converter 06.pdf

### TOPICS IN THIS LECTURE (7)

☐ Ćuk circuit: energy transfer through C1; output inverted
☐ $V_o = -V_s D/(1 - D)$ (from average capacitor voltage $V_{C1} = V_s - V_o$)
☐ Continuous input and output currents (low ripple, low EMI)
☐ Inductor ripples $\Delta i_{L1} = V_sDT/L1$, $\Delta i_{L2} = V_sDT/L2$
☐ C1 ripple $\Delta v_{C1} = V_oD/(RC1f)$; output ripple $\Delta V_o/V_o = (1 - D)/(8L2C2f^2)$

### DO WITH YOUR OWN HAND

☐ Ex 6-7: choose D = 0.6, f, L1, L2, C1, C2
☐ Write the advantages of Ćuk over buck-boost (2024 Q6b, 6 marks)

### PAST-PAPER HITS

- 2024 Q6b (advantage of Ćuk over buck-boost)
- 2023 Q4d (SEPIC vs Ćuk)

### FORMULA SHEET

- $V_o = -V_sD/(1 - D)$
- $\Delta v_{C1} = V_oD/(RC1f)$

☐ L_min for continuous current in L1 and L2
☐ Example 6-7: Ćuk design (12 V to 18 V, 40 W)

- $\Delta V_o/V_o = (1 - D)/(8L_2C_2f^2)$
- $L_{1,min} = (1 - D)^2R/(2Df)$; $L_{2,min} = (1 - D)R/(2f)$

### HOW TO STUDY IT

1. Learn the circuit and the 'why': both inductors in series with input and output make both currents continuous. Do Ex 6-7 once.

### CLOSED-BOOK SELF-TEST (ANSWER WITHOUT LOOKING)

☐ Draw a Ćuk converter. What is its advantage over a buck-boost?

## L15 DC-DC Converter 07: SEPIC and Interleaved Converters

Part B · ★★★ Must master · 0.75 h · Day 2 · Hart 6.8, 6.9
Slide: Lecture 15 DC-DC Converter 07.pdf

### TOPICS IN THIS LECTURE (7)

☐ SEPIC circuit (L1, C1, L2, switch, diode, C2); output not inverted
☐ $V_o = V_s D/(1 - D)$; $V_{C1} = V_s$
☐ Inductor currents: $I_{L1} = I_s$, $I_{L2} = I_o$; ripples $\Delta i_{L1} = \Delta i_{L2} = V_{sDT/L}$
☐ Capacitor ripples $\Delta v_{C1} = V_oD/(RC_1f)$, $\Delta V_o/V_o = D/(RC_2f)$
☐ Example 6-8: SEPIC analysis
☐ SEPIC vs buck-boost and vs Ćuk
☐ Interleaved converters: phase-shifted parallel stages, reduced ripple

### DO WITH YOUR OWN HAND

☐ Ex 6-8 (9 V, D = 0.4, 100 kHz, L = 90 µH, C = 80 µF, $I_o$ = 2 A): $V_o$ = 6 V, $I_{L1}$ = 1.33 A, $I_{L2}$ = 2 A, $\Delta i_L$ = 0.4 A, $\Delta v_C$ = 0.1 V
☐ Table: SEPIC vs Ćuk vs buck-boost (polarity, input current, ground, components)

### PAST-PAPER HITS

- 2021 Q4d (draw SEPIC, advantage over buck-boost)
- 2023 Q4d (SEPIC vs Ćuk)
- 2024 Q5a (what makes SEPIC superior to buck-boost, 6 marks)

### FORMULA SHEET

- $V_o = V_sD/(1 - D)$
- $\Delta i_{L1} = \Delta i_{L2} = V_{sDT/L}$
- $\Delta V_o/V_o = D/(RC_2f)$

### HOW TO STUDY IT

1. Learn the circuit and a 4-row comparison table. SEPIC: non-inverting, continuous input current, common ground, works as buck or boost.

### CLOSED-BOOK SELF-TEST (ANSWER WITHOUT LOOKING)

☐ What makes SEPIC superior to buck-boost?
☐ Solve Ex 6-8

## L23 Resonant Converter 01: Zero-Current Switching (ZCS)

Part B · ★★ Important · 0.75 h · Day 3 · Hart 9.1, 9.2
Slide: Lecture 23 Resonant Converter 01.pdf

### TOPICS IN THIS LECTURE (9)

☐ Why resonant converters: switching losses rise with frequency; switch at zero current or voltage
☐ ZCS resonant switch converter circuit ($L_r$, $C_r$, switch, diode, $L_o$, $C_o$)
☐ Mode 1 (0 to t1): $i_L$ rises linearly, $t_1 = I_oL_r/V_s$
☐ Mode 2 (t1 to t2): resonance, $i_L = I_o + (V_s/Z_0)\sin \omega_0(t - t_1)$, switch current returns to zero
☐ Mode 3 (t2 to t3): $C_r$ discharges at constant $I_o$

### DO WITH YOUR OWN HAND

☐ Ex 9-1 ($V_s$ = 12 V, $C_r$ = 0.1 µF, $L_r$ = 10 µH, $I_o$ = 1 A): $Z_0$ = 10 Ω, peak $i_L$ = 2.2 A, find $V_o$ and $f_{s,max}$
☐ Draw the four-mode waveforms ($i_{Lr}$, $v_{Cr}$)

### PAST-PAPER HITS

- 2021 Q6d (ZVS, with equations provided)
- Resonant topic: 2021, 2022, 2024

### FORMULA SHEET

- $\omega_0 = 1/\sqrt{L_rC_r}$, $Z_0 = \sqrt{L_r/C_r}$
- $t_1 = I_oL_r/V_s$

☐ Mode 4 (t3 to T): diode freewheels
☐ Condition for ZCS: $I_o < V_s/Z_0$; $Z_0 = \sqrt{(L_r/C_r)}$, $\omega_0 = 1/\sqrt{(L_rC_r)}$
☐ Output voltage $V_o = V_s f_s [t1/2 + (t2 - t1) + (t3 - t2)]$; $V_o$ proportional to $f_s$
☐ Example 9-1

- $I_{L,max} = I_o + V_s/Z_0$
- ZCS needs $I_o < V_s/Z_0$

### HOW TO STUDY IT

1. Learn the circuit and 4 modes as a table (interval, what conducts, equation). In exams the long equations are usually given; you must know how to use them.

### CLOSED-BOOK SELF-TEST (ANSWER WITHOUT LOOKING)

☐ Explain the operation of a ZCS resonant converter with waveforms
☐ Solve Ex 9-1

## L24 Resonant Converter 02: Zero-Voltage Switching (ZVS) and Series Resonant Inverter

Part B · ★★ Important · 0.75 h · Day 3 · Hart 9.3, 9.4
Slide: Lecture 24 Resonant Converter 02.pdf

### TOPICS IN THIS LECTURE (8)

☐ ZVS converter circuit ($C_r$ across the switch, $L_r$ in series)
☐ Mode 1: switch opens, $C_r$ charges linearly, $t1 = V_sC_r/I_o$
☐ Mode 2: resonance, $v_C = V_s + I_oZ_0 \sin \omega_0(t - t1)$; requires $I_oZ_0 > V_s$
☐ Mode 3: diode across switch conducts; switch turns on at zero voltage
☐ Mode 4: switch on
☐ Output voltage $V_o = V_s[1 - f_s(t3 - t1/2)]$
☐ Example 9-2 ($V_s = 20$ V, $L_r = 1$ µH, $C_r = 0.047$ µF, $I_o = 5$ A): $f_s$ for $V_o = 10$ V
☐ Series resonant inverter introduction

### DO WITH YOUR OWN HAND

☐ Ex 9-2
☐ 2021 Q6d ($V_s = 5$ V, $I_o = 3$ A, $L_r = 1$ µH, $C_r = 0.01$ µF: $V_o$ at 500 kHz; $f_s$ for $V_o = 2.5$ V)

### PAST-PAPER HITS

- 2021 Q6d (ZVS numerical, 6 marks)

### FORMULA SHEET

- $t1 = V_sC_r/I_o$
- $t2 = (1/\omega_0)\sin^{-1}(-V_s/(I_oZ_0)) + t1$
- $V_o = V_s[1 - f_s(t3 - t1/2)]$
- ZVS needs $I_oZ_0 > V_s$

### HOW TO STUDY IT

1. Same table method as ZCS. Do the 2021 Q6d numerical since the equations were given in the paper.

### CLOSED-BOOK SELF-TEST (ANSWER WITHOUT LOOKING)

☐ Explain ZVS operation and its condition
☐ Solve 2021 Q6d

## L25 Inverter 01: Half-Bridge Inverter, Fourier Analysis, Performance Parameters

Part B · ★★★ Must master · 1.25 h · Day 3 · Rashid 6.4 (class notes)
Slide: Lecture 25 Inverter 01.pdf (handwritten notes)

### TOPICS IN THIS LECTURE (12)

☐ Inverter: DC to AC; VSI, CSI, variable DC-link inverter
☐ Half-bridge: two switches, two capacitors ($V_{dc}/2$ each); output $\pm V_{dc}/2$
☐ Switch states and output waveform; $V_{o,rms} = V_{dc}/2$

### DO WITH YOUR OWN HAND

☐ Derive the Fourier series of the half-bridge output
☐ Class example ($V_{dc} = 48$ V, $R = 2.4$ Ω): $V_{o1} = 21.6$ V, $P = 240$ W, $I_p = 10$ A, $I_{avg(switch)} = 5$ A, $V_{BR} = 48$ V, $I_s = 5$ A, THD = 48.34%, DF = 3.8%, $HF_3 = 33.3\%$, LOH = 3rd (7.2 V)

☐ Fourier series: $b_n = 2V_{dc}/(n\pi)$ for odd n; $v_o = \Sigma\ (2V_{dc}/n\pi) \sin n\omega t$
☐ Fundamental: $V_{o1} = 2V_{dc}/(\sqrt{2}\pi) = 0.45V_{dc}$
☐ Harmonic factor $HF_n = V_{on}/V_{o1}$
☐ THD = $(1/V_{o1})\sqrt{(V_o^2 - V_{o1}^2)} = 48.34\%$
☐ Distortion factor DF = $(1/V_{o1})\sqrt{(\Sigma(V_{on}/n^2)^2)}$; lowest-order harmonic (LOH)
☐ Example: $V_{dc}$ = 48 V, R = 2.4 Ω
☐ Gating sequence; gate isolation (optocoupler, pulse transformer)
☐ Feedback diodes for reactive loads
☐ RL load: $i_o = \Sigma\ (2V_{dc}/(n\pi Z_n)) \sin(n\omega t - \theta_n)$; $P = I_{o1,rms}^2 R$ (approximate)

### PAST-PAPER HITS

- 2022 Q5b (principle of a square-wave inverter, 8 marks)
- 2023 Q5d (THD of a square-wave inverter)
- 2022 Q1a(i) (short note on THD)

### FORMULA SHEET

- $V_{o,rms} = V_{dc}/2$
- $v_o = \Sigma\ (2V_{dc}/n\pi) \sin n\omega t$ (n odd)
- $V_{o1} = 0.45V_{dc}$
- $THD = \sqrt{(V_o^2 - V_{o1}^2)}/V_{o1}$
- $DF = (1/V_{o1})\sqrt{\Sigma(V_{on}/n^2)^2}$
- $HF_n = V_{on}/V_{o1}$

### HOW TO STUDY IT

1. These are handwritten notes, so read them slowly and rewrite each derivation neatly in your own copy.
2. Learn the 4 definitions (HF, THD, DF, LOH) word for word.
3. Solve the 48 V, 2.4 Ω example completely. Inverters were barely asked in 2024 but were your batch's Assessment #5 and #6 topic, so expect a full question here.

### CLOSED-BOOK SELF-TEST (ANSWER WITHOUT LOOKING)

☐ Derive the Fourier series of a half-bridge inverter output
☐ Define THD, DF, HF and LOH
☐ Solve the 48 V, 2.4 Ω example

## L26 Inverter 02: Single-Phase Full-Bridge Inverter (R, L and RL loads)

Part B · ★★★ Must master · 1 h · Day 3 · Rashid 6.5 / Hart 8.2 to 8.4 (class notes)
Slide: Lecture 26 Inverter 02.pdf (handwritten notes)

### TOPICS IN THIS LECTURE (8)

☐ Full-bridge: 4 switches; S1, S2 on gives $+V_{dc}$; S3, S4 on gives $-V_{dc}$
☐ $V_{o,rms} = V_{dc}$
☐ Fourier series: $v_o = \Sigma\ (4V_{dc}/n\pi) \sin n\omega t$ (n odd)
☐ $V_{o1} = 4V_{dc}/(\sqrt{2}\pi) = 0.90V_{dc}$; HF, DF, THD = 48.34%
☐ Inductive load: conduction sequence of switches and feedback diodes (D1 to D4)
☐ RL load current: $i_o = \Sigma\ (4V_{dc}/(n\pi Z_n)) \sin(n\omega t - \theta_n)$, $Z_n = \sqrt{(R^2 + (n\omega L)^2)}$
☐ Power $P = \Sigma\ I_{n,rms}^2 R$; DC source current $I_s = P/V_{dc}$
☐ Amplitude and harmonic control (quasi-square wave with dead angle)

### DO WITH YOUR OWN HAND

☐ Derive the full-bridge Fourier series and $V_{o1} = 0.90V_{dc}$
☐ RL-load square-wave inverter ($V_{dc}$ = 100 V, R = 10 Ω, L = 25 mH, 60 Hz): Fourier current, P, average source current
☐ Show the feedback-diode conduction intervals for an inductive load

### PAST-PAPER HITS

- 2021 Q6a and 2022 Q5c (full-bridge square wave, RL load: same numbers both years)
- 2022 Q5d (how to control amplitude and harmonics, 7 marks)
- 2021 Q6b (switching diagram to reduce 3rd harmonic)
- 2019 TT-2 (fifth-harmonic elimination; α for 90 V fundamental)

### FORMULA SHEET

- $v_o = \Sigma\ (4V_{dc}/n\pi) \sin n\omega t$
- $V_{o1,rms} = 0.90V_{dc}$
- $I_n = V_n/\sqrt{(R^2 + (n\omega L)^2)}$
- $P = \Sigma\ (I_{n,max}/\sqrt{2})^2 R$
- $I_s = P/V_{dc}$

### HOW TO STUDY IT

1. Write the Fourier derivation once for full-bridge (it is the half-bridge result with V_dc/2 replaced by V_dc).
2. Solve the 100 V, 10 Ω, 25 mH problem with the Fourier method: compute I_1, I_3, I_5, I_7 and add the powers. It appeared twice.

### CLOSED-BOOK SELF-TEST (ANSWER WITHOUT LOOKING)

☐ Derive the output Fourier series of a full-bridge inverter
☐ Solve the 100 V, 10 Ω, 25 mH RL-load problem
☐ How are amplitude and harmonics controlled in an inverter?

## L27 Problem Solving Class 01 = Assessment #5: Quasi-Square-Wave Full-Bridge Inverter

Part B · ★★★ Must master · 0.5 h · Day 3 · Your batch's Assessment #5
Slide: Lecture 27 Problem Solving Class 01.pdf

### TOPICS IN THIS LECTURE (5)

☐ Q1, Q2 conduct from δ to π − δ; Q3, Q4 from π + δ to 2π − δ (quasi-square wave)
☐ RMS output: $V_{o,rms} = V_{dc}\sqrt{((\pi - 2\delta)/\pi)}$
☐ Fourier series: $V_n = (4V_{dc}/n\pi)\cos(n\delta)$, n odd
☐ Fundamental rms and THD as functions of δ
☐ δ = 30° removes the 3rd harmonic (harmonic elimination)

### DO WITH YOUR OWN HAND

☐ δ = π/6: $V_{o,rms} = 0.8165V_{dc}$, $V_{o1,rms} = 0.7797V_{dc}$, THD = 31.1%
☐ Sketch the quasi-square output waveform with δ marked

### PAST-PAPER HITS

- Your own Assessment #5 (very likely to return as a final question)
- 2022 Q5d, 2021 Q6b, 2019 TT-2 (amplitude and harmonic control)

### FORMULA SHEET

- $V_{o,rms} = V_{dc}\sqrt{(1 - 2\delta/\pi)}$
- $V_n = (4V_{dc}/n\pi)\cos n\delta$
- $THD = \sqrt{(V_o^2 - V_{o1}^2)/V_{o1}}$

### HOW TO STUDY IT

1. Solve the assessment again from scratch under a 15-minute timer.

### CLOSED-BOOK SELF-TEST (ANSWER WITHOUT LOOKING)

☐ Solve Assessment #5 (both questions) in 15 minutes

## L28 Inverter 03: Three-Phase Inverter (180° conduction)

Part B · ★★★ Must master · 1.25 h · Day 3 · Rashid 6.8 (class notes)
Slide: Lecture 28 Inverter 03.pdf (handwritten notes)

### TOPICS IN THIS LECTURE (9)

☐ Three-phase bridge: six switches S1 to S6, gated 60° apart, each conducts 180°
☐ Gate sequence table; three switches on at a time (e.g. 5, 6, 1 then 6, 1, 2 ...)
☐ Mode equivalent circuits with Y-connected R load: $v_{an} = V_{dc}/3$ or $2V_{dc}/3$
☐ Phase voltage six-step waveform; line voltage quasi-square (120° wide)
☐ Line rms: $V_L = \sqrt{(2/3)}V_{dc} = 0.8165V_{dc}$
☐ Line Fourier series: $v_{ab} = \Sigma (4V_{dc}/n\pi)\cos(n\pi/6) \sin n(\omega t + \pi/6)$; triplen harmonics absent
☐ $V_{L1} = 0.7797V_{dc}$; phase rms $V_p = V_L/\sqrt{3} = 0.4714V_{dc}$

### DO WITH YOUR OWN HAND

☐ Fill the 180° gating table and draw $v_{ab}$, $v_{bc}$, $v_{ca}$, $v_{an}$
☐ Derive $V_{L,rms} = 0.8165V_{dc}$ and the Fourier series of $v_{ab}$

### PAST-PAPER HITS

- Not asked 2021 to 2024 in finals; your batch had the 120° version as Assessment #6, so the 180° version is the natural final question

### FORMULA SHEET

- $V_L = 0.8165V_{dc}$
- $V_{L1} = 0.7797V_{dc}$
- $V_p = 0.4714V_{dc}$
- $v_{ab} = \Sigma (4V_{dc}/n\pi)\cos(n\pi/6) \sin n(\omega t + \pi/6)$

☐ Phase Fourier series v_an (shifted by −π/6)
☐ RL load current per harmonic

### HOW TO STUDY IT

1. Draw the gate table and waveform grid by yourself twice (copying the notes). Then do the rms integral: $V_L^2 = (1/\pi)\int$ from 0 to 2π/3 of $V_{dc}^2$ d(ωt).

### CLOSED-BOOK SELF-TEST (ANSWER WITHOUT LOOKING)

☐ Draw the 180° gating sequence and phase/line voltages of a 3-phase inverter
☐ Derive V_L,rms and the Fourier series of the line voltage

## L29 Inverter 04: Multilevel Inverters (Cascaded H-Bridge)

Part B · ★★★ Must master · 1 h · Day 3 · Class notes
Slide: Lecture 29 Inverter 04.pdf (handwritten notes)

### TOPICS IN THIS LECTURE (8)

☐ Why multilevel: more sinusoidal output, lower THD, lower dv/dt, higher voltage with low-rated switches
☐ Three-level H-bridge: +V_dc, 0, −V_dc and the switch pairs for each
☐ Five-level cascaded inverter (2 H-bridges): +2V_dc, +V_dc, 0, −V_dc, −2V_dc
☐ Seven-level cascaded inverter (3 H-bridges, 12 switches): switch combinations for each level
☐ Levels = 2s + 1 (s = number of separate DC sources / H-bridges)
☐ Staircase waveform and its rms: V_o,rms = √2V_dc (7-level example)
☐ Fundamental V_o1 ≈ 1.35V_dc (rms) and THD ≈ 31.5% (vs 48.34% for 2-level)
☐ Pattern swapping: equal power sharing among sources

### DO WITH YOUR OWN HAND

☐ Draw a 5-level cascaded inverter with its switching table and output waveform (2024 Q5c)
☐ Draw the 7-level circuit and list switches for each level
☐ Explain why pattern swapping is important (2023 Q5b)

### PAST-PAPER HITS

- 2024 Q5c (design a 5-level inverter, 5 marks)
- 2023 Q5b (3-level cascade multi-source, why pattern swapping)
- 2021 Q5b (four-source cascade, equalisation of source power)

### FORMULA SHEET

- `Levels = 2s + 1`
- `THD falls as levels increase`

### HOW TO STUDY IT

1. Draw one H-bridge and its 3 levels; then stack two (5-level) and three (7-level). Write the switching table. Learn the 3-line pattern-swapping explanation.

### CLOSED-BOOK SELF-TEST (ANSWER WITHOUT LOOKING)

☐ Design a 5-level inverter and explain its operation
☐ Why is pattern swapping needed in a multi-source inverter?

## L30 Problem Solving Class 02 = Assessment #6: Three-Phase Inverter, 120° Conduction

Part B · ★★★ Must master · 0.5 h · Day 3 · Your batch's Assessment #6
Slide: Lecture 30 Problem Solving Class 02.pdf

### TOPICS IN THIS LECTURE (4)

☐ 120° conduction: only two switches on at a time; each phase is +, floating, −, floating
☐ Phase voltage: ±V_dc/2 for 120°, 0 for 60°; V_p,rms = V_dc/√6 = 0.408V_dc
☐ Line voltage levels V_dc/2, V_dc, V_dc/2; V_L,rms = V_dc/√2 = 0.7071V_dc
☐ Comparison with 180° mode

### DO WITH YOUR OWN HAND

☐ Derive V_L,rms = V_dc/√2 for the 120° mode (with waveform)

### PAST-PAPER HITS

- Your own Assessment #6 (10 marks)

### FORMULA SHEET

- `V_p = 0.4082V_dc`
- `V_L = 0.7071V_dc`

### HOW TO STUDY IT

1. Draw the 120° gate table, then v_an, then v_ab, then integrate. 15-minute timed attempt.

### CLOSED-BOOK SELF-TEST (ANSWER WITHOUT LOOKING)

☐ Derive the line-to-line rms voltage of a 3-phase inverter in 120° mode

## L31_32 AC-AC Converter 01-02: Single-Phase and Three-Phase AC Voltage Controllers

Part B · ★★★ Must master · 1.5 h · Day 4 · Rashid ch.11 (class notes)
Slide: Lecture 31_32 AC-AC Converter 01_02.pdf (handwritten notes)

### TOPICS IN THIS LECTURE (11)

☐ AC-AC conversion: change rms voltage (AC voltage controller) or frequency (cycloconverter)
☐ Single-phase full-wave controller: two antiparallel SCRs (or a TRIAC) with R load
☐ Output waveform for delay α; derivation $V_o = (V_m/\sqrt{2})\sqrt{((1/\pi)(\pi - \alpha + \sin 2\alpha/2))}$
☐ Input power factor $PF = P/(V_s I_s) = V_o/V_s$
☐ Thyristor average and rms current derivations
☐ Class example: 120 V, 60 Hz, R = 10 Ω, α = 90°
☐ Three-phase bidirectional (6 SCRs) controller with Y-connected R load
☐ Gate pulses for α = 60° on the 3-phase grid; modes with 3 or 2 SCRs conducting
☐ Phase output v_an segments: v_an, v_ab/2, v_ac/2, 0 for each 60° interval
☐ RMS of v_an by piecewise integration
☐ Applications: heating, lighting, soft starters, induction motor speed control

### DO WITH YOUR OWN HAND

☐ Class example (120 V, R = 10 Ω, α = 90°): $V_o$ = 84.85 V, P = 720 W, PF = 0.707, $I_{T,avg}$ = 2.7 A, $I_{T,rms}$ = 6 A
☐ Derive $V_{o,rms}$ for the single-phase controller
☐ Draw the 3-phase controller circuit and the v_an waveform for α = 60°

### PAST-PAPER HITS

- 2024 Q5d (single-phase controller α = 90°, 120 V: class example verbatim, 8 marks)
- 2024 Q6a (3-phase bidirectional controller α = 60° waveforms, 8 marks)
- 2022 Q5a (briefly explain AC voltage controllers)
- Your Assessment #7

### FORMULA SHEET

- $V_o = V_s\sqrt{(1 - \alpha/\pi + \sin 2\alpha/2\pi)}$
- $PF = V_o/V_s$
- $I_{T,avg} = (\sqrt{2}V_s/2\pi R)(1 + \cos\alpha)$
- $I_{T,rms} = I_o/\sqrt{2}$

### HOW TO STUDY IT

1. The 2024 paper used both class-note examples from this lecture. Treat every drawing here as an exam question.
2. Do the single-phase derivation and the α = 90° example, then copy the α = 60° 3-phase drawing until you can draw it from memory.

### CLOSED-BOOK SELF-TEST (ANSWER WITHOUT LOOKING)

☐ Derive the rms output of a single-phase AC voltage controller and solve for α = 90°
☐ Draw the 3-phase bidirectional controller and v_an for α = 60°

## L33 AC-AC Converter 03: Cycloconverters

Part B · ★★★ Must master · 0.75 h · Day 4 · Rashid 11.x (class notes)
Slide: Lecture 33 AC-AC Converter 03.pdf (handwritten notes)

### TOPICS IN THIS LECTURE (7)

☐ Cycloconverter: direct AC to AC at a lower frequency (no DC link); vs rectifier + inverter
☐ Single-phase cycloconverter: P-converter and N-converter (back-to-back bridges)
☐ Operation for $f_o = f_s/3$ ($T_o = 3T_s$): P-converter on for $T_o/2$, N-converter for $T_o/2$
☐ Operation for $f_o = f_s/4$ and $f_o = f_s/5$

### DO WITH YOUR OWN HAND

☐ Draw the single-phase cycloconverter and the output waveform for $f_o = f_s/3$
☐ Write definition + 4 applications

### PAST-PAPER HITS

- Asked every year: 2021 Q3a, 2022 Q3a,b, 2023 Q6b, 2024 Q4a

### FORMULA SHEET

☐ Three-phase to single-phase cycloconverter (two 3-phase full converters), f_o = 12 Hz from 60 Hz
☐ Three-phase cycloconverter (three such groups)
☐ Applications: low-speed high-power AC drives (rolling mills, cement kilns, ship propulsion, mine hoists)

- `f_o = f_s/k; T_o = kT_s`

### HOW TO STUDY IT

1. Guaranteed short question. Learn the definition, circuit, f_s/3 waveform and applications. 45 minutes.

### CLOSED-BOOK SELF-TEST (ANSWER WITHOUT LOOKING)

☐ What is a cycloconverter? Explain a single-phase cycloconverter with waveforms; list applications

## L34 Problem Solving Class 03 = Assessment #7: Single-Phase AC Voltage Controller

Part B · ★★★ Must master · 0.25 h · Day 4 · Your batch's Assessment #7
Slide: Lecture 34 Problem Solving Class 03.pdf

### TOPICS IN THIS LECTURE (3)

☐ Single-phase full-wave controller, R = 10 Ω, 120 V, 60 Hz, asymmetric delay (T1 at 80°, T2 at 180° + 100°)
☐ RMS output with unequal delay angles
☐ Input power factor

### DO WITH YOUR OWN HAND

☐ $V_o^2 = (V_m^2/4\pi)[(\pi - \alpha_1 + \tfrac{1}{2}\sin 2\alpha_1) + (\pi - \alpha_2 + \tfrac{1}{2}\sin 2\alpha_2)]$: V_o = 84.85 V, PF = 0.707

### PAST-PAPER HITS

- Your own Assessment #7

### FORMULA SHEET

- `Asymmetric: average the two half-cycle integrals`

### HOW TO STUDY IT

1. Redo it in 10 minutes after L31_32.

### CLOSED-BOOK SELF-TEST (ANSWER WITHOUT LOOKING)

☐ Solve Assessment #7 in 10 minutes

## L35 Solar PV System and Course Wrap-Up

Part B · ★★ Important · 0.5 h · Day 4 · Slides + video
Slide: Lecture 35 Solar PV System.pdf

### TOPICS IN THIS LECTURE (5)

☐ How a solar system works (PV effect, panels, DC output)
☐ Stand-alone PV block diagram: PV array, charge controller / MPPT DC-DC converter, battery, inverter, AC loads, DC loads
☐ Function of each component
☐ DC-DC converters used: buck (battery charging), boost (raise PV voltage), buck-boost / SEPIC / Ćuk (MPPT)
☐ Course map: devices (L2-8), DC-DC (L9-16), rectifiers (L17-22), resonant (L23-24), inverters (L25-30), AC-AC (L31-34), renewables (L35)

### DO WITH YOUR OWN HAND

☐ Draw the stand-alone PV system diagram and write one line per block

### PAST-PAPER HITS

- 2024 Q6d (stand-alone PV schematic, 5 marks)
- 2021 Q5a and 2023 Q5a (DC-DC converter types used in stand-alone PV, 5 marks)

### HOW TO STUDY IT

1. Learn the block diagram and the converter list. 30 minutes.

**CLOSED-BOOK SELF-TEST (ANSWER WITHOUT LOOKING)**

☐ Draw the stand-alone solar PV system and describe its components
☐ Which DC-DC converters are used in a stand-alone PV system, and why?

# 6. External-examiner zone (not in the slides, but asked before)

Give these one session on Day 4. Even partial knowledge here can save 3 to 5 marks.

| Topic | What to know |
|---|---|
| **Parallel resonant DC-DC converter ($V_s = 100$ V, $L_r = 8$ µH, $C_r = 0.32$ µF, $R_L = 10$ Ω, $f_s = 120$ kHz)** | Asked in 2022 and 2024 with the same numbers but not in the slides. Method: $R_e = \pi^2 R_L/8 = 12.34$ Ω; fundamental $4V_s/\pi$; divide between $jX_L = j6.03$ Ω and $R_e \parallel (-j4.14$ Ω) to get $V_{C1,peak} \approx 190$ V; $V_o = 2V_{C1}/\pi \approx 121$ V. Check with Hart 9.6. |
| **Power computation with non-sinusoidal waveforms (Hart ch.2)** | Every TT-1 plus 2021 Q1c and 2023 Q1a,d. $P = V_0 I_0 + \Sigma V_{n,rms} I_{n,rms} \cos(\theta_n - \varphi_n)$; $PF = P/(V_{rms} I_{rms})$; $DF = I_1/I_{rms} = 1/\sqrt{(1 + THD^2)}$ (derivation asked); $THD = \sqrt{(I_{rms}^2 - I_1^2)}/I_1$; crest factor = $I_{peak}/I_{rms}$. |
| **Bipolar PWM inverter and harmonic elimination** | 2021 Q5c and 2023 Q5c (with a coefficient table given). $V_1 = m_a V_{dc}$; harmonics at $m_f$ and $m_f \pm 2$; $I_n = V_n/Z_n$; THD from the table. |
| **DC motor and induction motor speed control** | 2021 Q6c (separately excited DC motor drive), 2022 Q3c (induction motor), 2020 TT-2 (field chopping). Learn one circuit each and the basic equations $V_a = E_a + I_a R_a$, $E_a = K\omega$. |
| **Buck converter design (5 V to 3.3 V, 1 to 2 A, 2% ripple, $\Delta i_L \le 25\%$)** | 2022 Q2c. Use L09 to L12 formulas: $D = 0.66$; size L for the minimum current; C and ESR from the ripple limit. |
| **Inverter for a resistive load with THD ≤ 5% (LC filter design)** | 2023 Q6a. Use a square-wave inverter plus an LC filter; size the filter so the 3rd harmonic is reduced enough. |
| **TRIAC/DIAC structure, GTO, MCT; HVDC advantages** | 2023 Q6c, 2021 TT-1, 2023 Q3c. Quick notes are enough. |

# 7. In the exam hall

1. Read the whole paper first (3 minutes). Choose the optional question in each part by asking which one has more numericals you have practised.
2. Time: about 2 minutes per mark. A 10-mark question gets about 20 minutes. Keep 10 minutes at the end to review.
3. Start each part with your strongest compulsory question.
4. Derivations: circuit, then equivalent circuits, then aligned waveforms, then equations, then boxed result.
5. Numericals: formula, substitution, answer with unit, boxed. Partial marks depend on the formula line.
6. Compare / differentiate: always a table of at least 5 rows.
7. Surprise question: write the governing equation and draw the circuit anyway. Never leave a sub-part blank.

Numbers in this guide were checked against the solutions printed in the slides where available; the parallel resonant answer is our own calculation (the topic is not in the slides).